PROJECT PROPOSAL · VERSION 2.0 · AS BUILT, AND NEXT STEPS

Leads Scraper: Multi-Industry Lead Capture and Outreach Automation

What has been built, how a lead moves from capture to a conversation, what it costs to run in the US market, and what remains.

---

1. Executive summary

The original system found supplier businesses for one purpose (dropshipping) and contacted them by email, SMS and AI voice call. It has been rebuilt into a platform that can serve **any industry**: real estate, HVAC, AI automation services, or anything that sells to local businesses. The industry is chosen from the admin panel, and nothing industry-specific remains in the code.

What changed:

1. **Industry Profiles.** Search wording, scoring rules, offer, FAQ and follow-up sequence are saved as data. The original dropship setup is the first profile and scores leads exactly as before (checked against every combination of its rules).
2. **An Outreach Queue.** Hot and Warm leads join it automatically. Each lead follows its profile's sequence on its own clock, inside its own working hours (its time zone), with gaps and daily limits that you set.
3. **Email through Gmail and WhatsApp through a Chrome extension.** This removes the paid email platform and all SMS cost.
4. **Messages written by AI, approved by a person.** Each channel is Manual, Assisted or Auto. Nothing is sent without approval until a channel is switched to Auto.
5. **Replies are understood.** Replies are read, labelled, answered with a draft, and the lead's sequence stops. "STOP" or a refusal adds the lead to the do-not-contact list.
6. **Connection status on every page.** The admin panel shows whether WhatsApp Web is open and the email account works, warns when messages are waiting, and offers an Open WhatsApp Web button.

NOT BUILT YET

The voice agent and closing agent are deliberately last. The database is still SQLite; the move to Supabase is planned. A lead-capture key (Google Places) has not been added yet, so a first live batch has not been run.

HEADLINE COST (US MARKET, ESTIMATE)

Running cost is roughly **$0 to $60 per month** for 1,000 leads captured per month: capture, email and WhatsApp are free or nearly free, the AI costs a few dollars, and the only fixed cost is the database plan if Supabase Pro is chosen. Voice calls, when added, are the one real variable cost. Details are in section 9.

2. What the system does now

STAGE	WHAT HAPPENS	STATUS
Capture	Searches by the profile's wording and a city, filters by rating and reviews, saves each lead with its country and time zone. Source keys: Google Places, SerpAPI.	Built. Needs the Google Places key
Enrich and score	Crawls the website, evaluates the profile's rules (keywords, patterns, numbers, AI yes/no questions, behaviour), and gives a score, tier and an explanation of the score.	Built
Email check	Free check of syntax, throw-away domains and whether the domain accepts mail. No paid verification service.	Built
Outreach Queue	Hot and Warm leads (or Hot only) join automatically. Table with next step, due time, the lead's local time, tier and mode.	Built
Schedule	Per-lead follow-up timing, working hours in the lead's time zone, random gaps, daily limits, stop on reply.	Built
Write and approve	AI writes each message from the profile's offer. Review list: Approve, Edit, Rewrite, Skip.	Built
Send: email	Sent from the server through Gmail, threaded, with a STOP line and unsubscribe header. Replies and bounces are read from the inbox.	Built. Needs a Gmail App Password
Send: WhatsApp	A small Chrome extension sends from your own WhatsApp Web and reports replies back.	Built. First real send still to be tried
Replies	Labelled, answered with a draft, "not now" leads come back after a set time, opt-outs blocked.	Built
Open tracking	Optional invisible image; records which leads opened an email.	Built. Needs a public address and a switch in the settings file
Status and warnings	WhatsApp and Email chips in the top bar, a warning banner when something needs attention.	Built
Voice calls	Calls are done by hand and marked Done. The AI voice agent exists from the earlier version and is not connected to the new queue.	Later

The existing admin panel, its look, its pages and its charts are unchanged. New pages (Outreach queue, Industry profiles) use the same style. The public landing page and the login were removed, so the panel must be run on your own computer or a private network.

3. Multi-industry design

An *Industry Profile* is a saved set of settings chosen when a campaign starts:

- **Search wording**, for example "HVAC contractor {city}".
- **Scoring rules**, each worth points, in six kinds: a stored value (rating, reviews), keywords on the website, a pattern on the website, a yes/no question answered by AI, a built-in check, or the lead's behaviour after outreach (opened, replied, call connected). Penalties are negative points.
- **Offer**: what we sell, tone, language, a booking link and a list of questions and answers the AI may use.
- **Follow-up sequence**: the steps, the wait between them and which leads get each step.

Profiles can be created, edited, copied, archived, exported and imported from the admin panel. Leads keep the profile they were captured under, and scores record which profile version produced them. Data that differs per lead (what the website said, what the AI answered, CSV columns) is stored as flexible fields, so a new kind of client needs no new database column.

Scoring parity. The original dropshipping rules are expressed as the first profile, and an automated test compares it with the old fixed formula for 2,304 combinations of inputs. Points, tier and the "score incomplete" flag all match.

What limits a new industry is its data source. Google Places and SerpAPI suit local businesses. Companies or individual professionals need a CSV import or another data source.

4. Workflow: from capture to a conversation

1 CAPTURE Scrape by the profile's wording and a city.	**2 ENRICH** Crawl the site, run the profile's rules.	**3 SCORE** Hot, Warm or Cold, with the reasons.	**4 QUEUE** Hot and Warm leads join automatically.
5 WRITE AI writes each due step.	**6 APPROVE** You approve, or the channel is Auto.	**7 SEND** At a planned time inside the lead's hours.	

Default sequence (editable per profile)

DAY	STEP	CHANNEL	WHO GETS IT
0	First email	Email	Everyone
3	Follow-up email	Email	Everyone
4	WhatsApp message	WhatsApp	Hot and Warm
7	New-angle email	Email	Everyone
9	Call (done by hand for now)	Phone	Hot and Warm
14	Closing note	Email	Everyone

Each lead has its own clock

- A step's wait is counted from when the previous step was actually done for that lead, not from when leads were collected. Leads captured on different days follow different schedules.
- The due time is moved into the lead's own working hours in its own time zone (for example 09:00 to 17:00 on weekdays in Miami or Los Angeles).
- Between two sends on a channel there is a random gap inside the range you set, and each channel has a daily limit. Follow-ups are served before first contacts.
- A step the lead cannot receive (no email, wrong tier) is skipped and logged. A lead with only a phone number starts at the WhatsApp step.
- A reply, an opt-out or a bounce stops the sequence.

5. Automation and approval

Each channel has a mode, set on the Outreach queue page under Sending rules:

MODE	WHAT HAPPENS WHEN A STEP IS DUE
Manual	The system only shows it is due. You do it yourself and press Done.
Assisted (default)	The AI writes the message and it waits for your approval. You can edit it, have it rewritten, or skip the step.
Auto	The message is approved by itself and sent at its planned time, within the limits.

Preparing the data (capture, enrichment, scoring, queueing) is always automatic. Sending starts in Assisted mode, and the supervisor decides when each channel moves to Auto after the pilot: no bans or spam problems, replies read correctly, opt-outs tested, limits and cost tracking working. A separate switch decides whether the AI's answers to replies may also go out without approval; it is off by default.

---

6. Channels

6.1 Email: Gmail, not a paid platform

CRITERION	GMAIL WITH AN APP PASSWORD (BUILT)	INSTANTLY (BEFORE)
Monthly cost	$0	$47 to $97
Setup	Two-step verification, an App Password, IMAP on. About ten minutes, no Google Cloud project	Account, API key, campaign setup
Replies	Read from the inbox every few minutes	Webhook
Open tracking	Optional, needs a public address	Built in
Warm-up and rotation	Manual: small volume first	Built in

Good delivery needs a separate sending account or domain, a warm-up of one to two weeks at a low volume, and a daily limit well below Gmail's own (the default is 40). Every email ends with a line asking the lead to reply STOP if it is not relevant, and carries a List-Unsubscribe header. Bounces mark the address invalid.

Open tracking is optional. It adds an invisible image to each email, so PUBLIC_BASE_URL must be an address that the lead's mail app can open (your domain, or a tunnel while testing), and EMAIL_OPEN_TRACKING must be switched on. An "open" is a hint, not proof, because some mail apps load images in advance and some never do. Loads within seconds of sending are ignored.

6.2 WhatsApp: a Chrome extension

- The extension runs inside your own WhatsApp Web. It asks the backend what to send, sends it with a short pause, and reports sent, failed or not on WhatsApp. When a lead writes back it tells the backend.

- The backend decides what, when, how many and whether approval is needed. The extension only delivers.
- **Chrome must be open with a WhatsApp Web tab** (it can be pinned or minimised). The admin panel shows a WhatsApp chip, warns when messages are waiting and the tab is not open, and has an Open WhatsApp Web button. The extension can also open the tab by itself when messages are waiting.
- If the tab is closed nothing is lost: messages wait and go out when it opens again.
- Automated messaging to people who have not asked for it can get a number restricted or banned. Start with a dedicated number, low limits and the gaps from Sending rules.
- The official WhatsApp Business API is not suitable for cold first contact: it requires the recipient's prior opt-in, and per-message fees apply.

SMS is no longer used.

6.3 Voice calls (later)

Calls need a phone number. Three ways work with Twilio: register an existing US number as a Verified Caller ID (outbound calls only), buy a Twilio number (outbound and inbound), or connect an existing carrier through a SIP trunk. A Twilio trial account can only call verified numbers, so it must be upgraded before calling real leads. The existing voice agent can be reused; the planned saving is to replace paid speech recognition and the paid voice with cheaper options after listening tests.

---

7. Replies and the closing agent

7.1 What happens when a lead answers

THE AI READS IT AS	WHAT THE SYSTEM DOES
Interested	Drafts an answer that explains the offer briefly and invites a short call, using the profile's booking link. The lead's sequence stops.
Has a question	Drafts an answer using only the profile's offer and FAQ. If the answer is not there it says it will confirm.
Not now	Drafts a short, polite acknowledgement and brings the lead back into the sequence after a set number of days (default 45). Replies from before that no longer stop it.
Stop	Adds the lead's email, phone and domain to the do-not-contact list. Nothing more is sent.
Other	Automatic replies, wrong person, unclear: nothing is sent.

Answer drafts appear in the same approval list, with the lead's message shown above. A channel in Manual mode gets no draft, because you reply yourself. Out-of-office replies are ignored. A plain "STOP" is always honoured, even if the AI is unavailable.

7.2 Closing agent (later)

Level 1: qualify and book

A voice agent confirms interest, asks a few qualifying questions, answers common questions and books a meeting in the team calendar, then writes a summary to the CRM. A person closes the deal in the meeting.

Recommended.

Level 2: full closing

The agent states pricing and agrees terms on the call.

Not recommended yet. It may promise things incorrectly and larger deals usually need a person.

8. Safety and compliance

- **Nothing is sent without approval** until a channel is switched to Auto.
- **Do-not-contact list** checked for every lead; email, phone and website domain are blocked together. Opt-out keywords, a bare "STOP", or an AI-read refusal all add a lead.
- **Working hours** in the lead's own time zone, random gaps, and daily limits per channel.
- **Visible status:** a warning appears if messages are waiting and the channel cannot send them.

- **No login:** the panel and its API are open to anyone who can reach the server, so it must run on your own computer or a private network. If it is ever put on the internet, a login must be added first.
- **Legal:** in the US, automated and AI-voice calls and texts are regulated (TCPA, FCC rules) and cold email must follow CAN-SPAM. The software provides the controls above; the business remains responsible for lawful use. Legal review of the scripts is recommended before the pilot.

---

9. Costing (US market)

HOW TO READ THIS SECTION

Monthly estimates for the US market. Items marked with a dagger (†) come from memory or secondary sources and should be confirmed on the vendor's pricing page before budgets are fixed. Other figures come from the pages listed in the Sources section.

9.1 Monthly total

Assumed volume: 1,000 leads captured, 500 emailed, 150 contacted on WhatsApp.

LINE	MONTHLY COST	NOTE
Lead capture (Google Places, SerpAPI)	$0 to $20	Free allowance covers about 1,000 leads. See 9.2
Email (Gmail)	$0	Instead of $47 to $97 for a campaign platform
WhatsApp (extension)	$0	Instead of SMS
AI writing, scoring questions and replies (Groq)	$1 to $5†	A fraction of a cent per message
Database (Supabase)	$0 or $25	Free for development and the pilot; Pro for continuous use. See 9.3
Backend hosting†	$0 to $10	Or run on your own computer
Total without voice	**$1 to $60**	**Depends on lead volume and the database plan**

Compared with the earlier design (paid campaign platform, paid SMS, paid voice for every message), the monthly running cost falls from roughly $160 to $250 to roughly $1 to $60. Voice

calls, when added, are extra: about $0.12 to $0.18 per minute in the current build, or a few dollars per month if speech recognition and the voice are replaced by free options. A US phone number is about $1 to $2 per month†, or free if an existing number is registered as a caller ID.

9.2 Google Maps (Places API)

Text Search is $32 per 1,000 calls after the first 5,000 free calls each month. The per-lead Details call (phone and website) dominates at about $20 per 1,000 beyond a small free allowance†. A lead with phone and website costs about **$0.02 to $0.025**.

LEADS PER MONTH	ESTIMATED GOOGLE COST
500	about $0
1,000	$0 to $5
5,000	$80 to $100
10,000	$180 to $220

A billing account with a card is required even for free use, so a budget alert and a daily quota must be set. The scraper uses the older ("legacy") Places endpoint, which new Google Cloud projects may not be able to enable; a move to Places API (New) may then be needed.

9.3 Database: Supabase

PLAN	PRICE	INCLUDES	FIT
Free	$0	500 MB, a project is paused after a week of inactivity	Development and the pilot. A paused database would stop the dispatcher
Pro	$25 per month	8 GB, no pause	Continuous use

Website text is stored for every lead, so the 500 MB limit should be checked against real data during the pilot.

10. Risks and mitigations

RISK	MITIGATION
WhatsApp number restricted or banned	Dedicated number, low daily limits, random gaps, working hours, varied text, a STOP line, Assisted mode first
WhatsApp Web changes its internals	The extension reports a clear error instead of failing silently; the page script needs an update. The first real send should go to your own second number
The extension needs Chrome and a WhatsApp Web tab open	Status chip, warning banner, Open WhatsApp Web button, optional automatic opening, messages wait safely
Email lands in spam	Separate sender, warm-up, a low daily limit, STOP line, bounce handling
Legal exposure (TCPA, CAN-SPAM)	Do-not-contact list, working hours, approval, legal review; calls only after interest
AI writes something wrong	Approval by default; answers use only the profile's offer and FAQ; AI replies without approval is off by default
AI mislabels a reply	Labels are shown next to each lead; a refusal is acted on only when the AI is clear; a plain STOP never depends on the AI
Open tracking counts scanners or misses opens	Treated as a hint; very early loads are ignored
Google Places legacy endpoint unavailable	Move to Places API (New); SerpAPI as an alternative
No login on the admin panel	Run locally or on a private network; add a login before any public hosting
Supabase Free pauses or fills	Free for the pilot only; Pro for continuous use

11. Roadmap and status

PHASE	WORK	STATUS
Foundation	Free email check, tests, README, safe logging, login removed	Done
Industry Profiles	Profile table and page, rules-as-data scoring, profile-driven scraper and prompts, original setup kept as the first profile	Done
Outreach Queue and schedule	Queue page, per-lead clock, working hours by time zone, gaps, limits, Manual / Assisted / Auto	Done
Email and WhatsApp	Gmail sending and reply reading, WhatsApp extension and its backend contract, message writing and approval, status chips and warnings	Done. The WhatsApp send is not yet tried on a real account
Replies	Labelling, answer drafts, "not now" revisit, opt-outs, open tracking	Done
Go-live set-up	Google Places key, Gmail App Password, WhatsApp number, public address (for open tracking), Supabase connection and migrations	Next
Pilot	A small batch in one industry in Assisted mode; measure replies, bans, spam signals and cost per lead	Next
Full automation	Switch channels to Auto one at a time after approval	After the pilot
Voice and closing agent	Connect the voice agent to the queue; qualify and book meetings (Level 1)	Later
Separate workspaces	Only if the platform is offered to other businesses: each with its own data, keys and login	Optional

12. What is needed

1. A Google Places key (billing account with a budget alert) so that leads can be captured.
2. A Gmail account for sending, with an App Password and IMAP on; ideally a separate one from the main business address.
3. A dedicated WhatsApp number, and a computer that can keep Chrome open during sending hours.

4. For open tracking: a public address for the backend and the tracking switch turned on (optional).
5. Approval of the Supabase plan: Free for the pilot, Pro afterwards.
6. The first industry and city for the pilot, and the scoring rules agreed for it.
7. A legal review of the scripts before the pilot.

---

13. Sources and assumptions

- Twilio SMS pricing, US
- Twilio voice and number pricing summary (third party)
- Twilio documentation on verified caller IDs
- Instantly pricing (third party)
- Google Places API pricing in 2026 (third party)
- Supabase pricing in 2026 (third party)
- WhatsApp Business Platform pricing (third party)

Assumptions. US market; the monthly volumes in section 9.1. Third-party price summaries are not vendor-published rates. Figures marked † are unverified. Prices change without notice.
